

UNIT I

Open source software (OSS) is software that is distributed with its source code, making it available for use, modification, and distribution with its original rights. Source code is the part of software that most computer users don't ever see; it's the code computer programmers manipulate to control how a program or application behaves. Programmers who have access to source code can change a program by adding to it, changing it, or fixing parts of it that aren't working properly. OSS typically includes a license that allows programmers to modify the software to best fit their needs and control how the software can be distributed.

What is the Open Source Initiative?

The Open Source Initiative (OSI) was created to promote and protect open source software and communities.ⁱⁱ In short, the OSI acts as a central informational and governing repository of open source software. It provides rules and guidelines for how to use and interact with OSS, as well as providing code licensing information, support, definitions, and general community collaboration to help make the use and treatment of open source understandable and ethical.ⁱⁱ

How does OSS work?

Open source code is usually stored in a public repository and shared publicly. Anyone can access the repository to use the code independently or contribute improvements to the design and functionality of the overall project.

OSS usually comes with a distribution license. This license includes terms that define how developers can use, study, modify, and most importantly, distribute the software.ⁱⁱⁱ According to the Synopsys [Black Duck® KnowledgeBase](#), five of the most popular licenses are:

- MIT License
- GNU General Public License (GPL) 2.0—this is more restrictive and requires that copies of modified code are made available for public use
- Apache License 2.0
- GNU General Public License (GPL) 3.0
- BSD License 2.0 (3-clause, New or Revised)—this is less restrictive^{iv}

When source code is changed, OSS must include what was altered as well as the methods involved. Depending on the license terms, the software resulting from these modifications may or may not be required to be made available for free.

What are some examples of OSS?

- GNU/Linux
- Mozilla Firefox
- VLC media player
- SugarCRM
- GIMP
- VNC

- Apache web server
- LibreOffice
- jQuery

What are the differences between open source and closed source software?

Factors	Open source	Closed source
Price	Available for nominal or zero licensing and usage charges.	Cost varies based upon the scale of the software.
Freedom to customize	Completely customizable but it depends on the open source license. Requires in-house expertise.	Change requests must be made to the company selling the software. This includes bug fixes, features, and enhancements.
User-friendliness	Typically less user-friendly, but it can depend on the goals of the project and those maintaining it.	Typically more user-friendly. As a for-profit product, adoptability and user experience are often key considerations.
After-sales support	Some very popular pieces of open source software (e.g., OSS distributed by Red Hat or SUSE) have plenty of support. Otherwise, users can find help through user forums and mailing lists.	Dedicated support teams are in place. The level of service available depends on the service-level agreement (SLA).
Security	Source code is open for review by anyone and everyone. There is a widespread theory that more eyes on the code makes it harder for bugs to survive. However, security bugs and flaws may still exist and pose significant risk.	The company distributing the software (i.e., software owner) guarantees a certain level of support, depending on the terms of the SLA. Because the source code is closed for review, there can be security issues. If issues are found, the software distributor is responsible for fixing them.
Vendor lock-in	No vendor lock-in due to the associated cost. Integration into systems may create technical dependency.	In most cases, large investments are made in proprietary software. Switching to a different vendor or to an open source solution can be costly.
Stability	This will depend on the current user base, the parties maintaining the software, and the number of years in the market.	Older, market-based solutions are more stable. New products have similar challenges as open source products. If a distributor discontinues an application, the customer may be out of luck.

Popularity	Some open source solutions are very popular and are even market leaders (e.g., Linux, Apache).	In some industries, proprietary software is more popular, especially if it has been in the market for many years.
Total cost of ownership (TCO)	TCO is lower and upfront due to minimal or no usage cost, and depends on the level of maintenance required.	TCO is much higher and depends on the size of the user base.
Community participation	The community participating in development, review, critique, and enhancement of the software is the essence of open source.	Closed community.
Interoperability with other open source software	This will depend on the level of maintenance and goals of the group, but it is typically better than closed source software.	This will depend on the development standards.
Tax calculation	Difficult due to undefined monetary value.	Definite.
Enhancements or new features	Can be developed by the user if needed.	Request must be made to the software owner.
Suitability for production environment	OSS might not be technically well-designed or tested in a large-scale production environment.	Most proprietary software goes through multiple rounds of testing. However, things can still go wrong when deployed in a production environment.
Financial institution considerations	The financial industry tends to avoid open source solutions. If used, a vetting process must take place.	Financial institutions prefer proprietary software.
Warranty	No warranty available.	Best for companies with security policies requiring a warranty and liability indemnity.

What are the advantages and disadvantages of open source software?

Advantages

- Open source software is free.

- Open source is flexible; developers can examine how the code works and freely make changes to dysfunctional or problematic aspects of the application to better fit their unique needs.
- Open source is stable; the source code is publicly distributed, so users can depend on it for their long-term projects since they know that the code's creators cannot simply discontinue the project or let it fall into disrepair.
- Open source fosters ingenuity; programmers can use pre-existing code to improve the software and even come up with their own innovations.
- Open source comes with a built-in community that continuously modifies and improves the source code.
- Open source provides great learning opportunities for new programmers.^v

Disadvantages

- Open source can be harder to use and adopt due to difficulty setting it up and the lack of friendly user interfaces.
- Open source can pose compatibility issues. When attempting to program proprietary hardware with OSS, there is often a need for specialized drivers that are typically only available from the hardware manufacturer.
- Open source software can pose liability issues. Unlike commercial software, which is fully controlled by the vendor, open source rarely contains any warranty, liability, or infringement indemnity protection. This leaves the consumer of the OSS responsible for maintaining compliance with legal obligations.
- Open source can incur unexpected costs in training users, importing data, and setting up required hardware.

Open Source Principles.

- Licencees are free to use Open Source Software for any purpose whatsoever.
- Licencees are free to make copies of open source software and to distribute them without payment of royalties to a licensor.
- Licencees are free to create derivative works of open source software and to distribute them without payment of royalties to a licensor.
- Licencees are free to access and use the source code of open source software
- Licencees are free to combine open source and other software.

Open Standard Requirements for software

- The criteria
- No intentional Secrets
- Availability
- Patents
- No Agreements
- No OSR-Incomplete Dependencies

Open Sources Successes

- More on server than client side
 - More expert users of servers
 - Server Operating systems
 - Linux

- Certain Server software
 - Web
 - Mail
 - DNS
- So far less so with other server software
 - Database(MySQL rising, less easy to measure)
- Software for technical users

What do you mean by free software?

Free software is **software that can be freely used, modified, and redistributed with only one restriction:** any redistributed version of the software must be distributed with the original terms of free use, modification, and distribution.

The most common is “Open Source Software”, which has since evolved to refer to a subtly different sense of freedom. Free software is also known as “Software libre”, “free libre”, and Open Source software” (“FOSS”), and “free and open source software (FOSS).

- The Freedom to run the program for any purpose
- The Freedom to study how the program works, and adapt it to your needs.
- The Freedom to redistribute copies so you help your neighbour.
- The freedom to improve the program, and release your improvements to the public, so that the whole community get benefits.

Free Software License Providers:

- GNU General Public Licence
- GNU Lesser General PublicLicence
- BSD Licences
- Mozilla Licences
- MIT Licences
- Apache Licence
- The “as-is” Release Model

Public Domain

Public domain comprises the body of knowledge and innovation (especially creative works such as writing, art, music and inventions) in relation to which no person or other legal entity can establish or maintain proprietary interests within a particular legal jurisdiction.

Public domain software:

Programs that are uncopyrighted because their authors intended to share them with everyone else are in the public domain. The UNIX community has developed a number of such programs over the years. Programs in the public domain can be used without restrictions as components of others programs.

Berkeley Software Distribution

BSD (originally: **Berkeley Software Distribution**) refers to the particular version of the UNIX operating system that was developed at and distributed from the University of California at Berkeley.

What is Free Software Foundation and explain the GNU project?

The Free Software Foundation (FSF) is a **nonprofit organization that promotes the creation, distribution and amendment of free software programs and applications without any restriction**. FSF was founded by Richard Stallman in 1985 as part of the free software movement and the earlier GNU Project.

Difference between Free Software and Open Source Software:

S.No.	Free Software	Open Source Software
1.	Software is an important part of people's lives.	Software is just software. There are no ethics associated directly to it.
2.	Software freedom translates to social freedom.	Ethics are to be associated to the people not to the software.
3.	Freedom is a value that is more important than any economical advantage.	Freedom is not an absolute concept. Freedom should be allowed, not imposed.
4.	Examples: The Free Software Directory maintains a large database of free-software packages. Some of the best-known examples include the Linux kernel, the BSD and Linux operating systems, the GNU Compiler Collection and C library; the MySQL relational database; the Apache web server; and the Sendmail mail transport agent.	Examples: Prime examples of open-source products are the Apache HTTP Server, the e-commerce platform osCommerce, internet browsers Mozilla Firefox and Chromium (the project where the vast majority of development of the freeware Google Chrome is done) and the full office suite LibreOffice.

UNIT II

Open Source Initiative(OSI)

Open source is a term describing a means of developing and distributing software that ensures software is available for use, modification, and redistribution by anyone. Generally, anyone can download open source software for free or a small fee, and can use, share, borrow, or change it without restriction. Open source practice promotes software reliability and quality by supporting independent peer review and rapid evolution of source code.

The Open Source Initiative (OSI) is a non-profit corporation whose goal is to promote the use of open source software in the commercial world. To accomplish this goal, OSI maintains and promotes the Open Source Definition and offers the OSI Certified Open Source Software Certification Mark and Program. To be OSI certified, the software must be distributed under a license that guarantees the right to read, redistribute, modify, and use the software freely. The Open Source Definition provided by OSI contains the following elements:

- Free redistribution
- Source code
- Derived works
- Integrity of the author's source code
- No discrimination against persons or groups
- No discrimination against fields of endeavor
- Distribution of license
- License must not be specific to a product
- License must not restrict other software
- License must be technology-neutral

You may be attracted to open source software for the following reasons:

- Rapid turnaround with regard to security patches

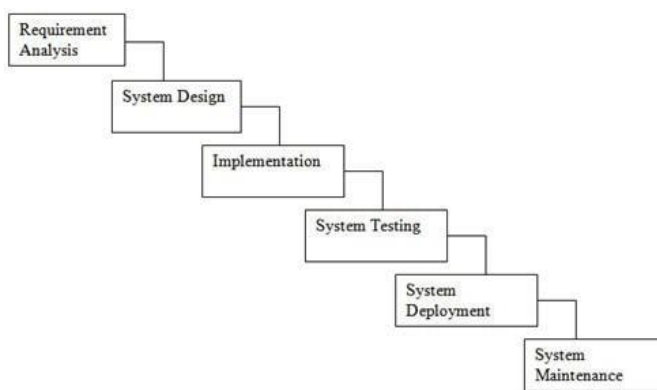
- Free availability
- Online access to software and source code without a large investment in time or money
- The opportunity to modify and improve source code

What is software methodology?

Software development methodology refers to structured processes involved when working on a project. It is a blend of design philosophies and pragmatic realism that stretches back to the early days of computing. The goal is to provide a systematic approach to software development.

WATER Fall Model

Waterfall model is an example of a Sequential model. In this model, the software development activity is divided into different phases and each phase consists of a series of tasks and has different objectives. In waterfall, the development of one phase starts only when the previous phase is complete.



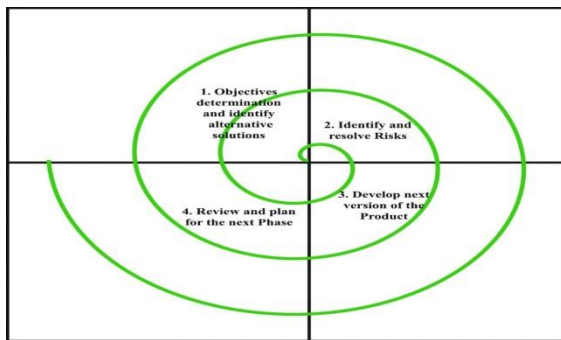
Waterfall Model - © www.SoftwareTestingHelp.com

Spiral Model

Spiral model is one of the most important Software Development Life Cycle models, which provides support for Risk Handling. In its diagrammatic representation, it looks like a spiral with many loops. The exact number of loops of the spiral is unknown and can vary from project to project. Each loop of the spiral is called a Phase of the software development

process. The exact number of phases needed to develop the product can be varied by the project manager depending upon the project risks. As the project manager dynamically determines the number of phases, so the project manager has an important role to develop a product using the spiral model.

The Radius of the spiral at any point represents the expenses(cost) of the project so far, and the angular dimension represents the progress made so far in the current phase.



What is the basic components of software freedom?

The freedom to run the program, for any purpose. The freedom to study how the program works, and modify it. The freedom to redistribute copies. The freedom to distribute copies of modified versions to others.

What is a software license?

A software license is a document that provides legally binding guidelines for the use and distribution of software.

Software licenses typically provide end users with the right to one or more copies of the software without violating copyrights. The license also defines the responsibilities of the parties entering into the license agreement and may impose restrictions on how the software can be used.

Software licensing terms and conditions usually include fair use of the software, the limitations of liability, warranties and disclaimers. They also specify protections if the software or its use infringes on the intellectual property rights of others.

Software licenses typically are proprietary, free or open source. The distinguishing feature is the terms under which users may redistribute or copy the software for future development or use.

How do software licenses work?

A software license is a document that states the rights of the developer and user of a piece of software. It defines how the software can be used and how it will be paid for. The following are some examples of specifications a license might include:

- how many times the software can be downloaded;
- what the software will cost; and
- what level of access users will have to the source code.

Licenses typically are expressed as an end-user license agreement or an enterprise license agreement. The license is a contract between the user or user organization and the developer. It defines the terms of the license. A user must agree to the terms of the license when acquiring the software.

Software also comes with a license key or product key. The key is used to identify and verify the specific version of the software. It is also used to activate the software on a specific computer or device.

What is copyright explain?

Copyright (or author's right) is a legal term used to describe the rights that creators have over their literary and artistic works. Works covered by copyright range from books, music, paintings, sculpture, and films, to computer programs, databases, advertisements, maps, and technical drawings.

What is the concept of copyleft?

Copyleft is the idea and the specific stipulation when distributing software that the user will be able to copy it freely, examine and modify the source code, and redistribute the software to

others (free or priced) as long as the redistributed software is also passed along with the copyleft stipulation.

What is copyleft and how is it different from copyright?

Copyright is the right that enable you to prevent unauthorized copying or selling of your work. Whereas Copyleft is a method using which you can modify the software or documentation and distribute it back to the open-source community.

Zero Marginal Cost

At the core of the financial aspects of Free and Open Source is the zero negligible expense of merchandise in an environment that is digital.

Internalization

Internalization occurs when a transaction is handled by an entity itself rather than routing it out to someone else. This process may apply to business and investment transactions, or to the corporate world. In business, internalization is a transaction conducted within a corporation rather than in the open market.

UNIT III

What is apache2 license?

The Apache License is **an open source software license released by the Apache Software Foundation (ASF)**. It's a popular and widely deployed license backed by a strong community. The Apache License allows you to freely use, modify, and distribute any Apache licensed product.

The Apache License (V2) - An Overview

The Apache License is recognised by the Open Source Initiative as a popular and widely deployed licence with a strong community. All of The Apache Software Foundation's projects, including the Apache HTTP Server project whose software powers more than half of the Internet's web servers, use this licence. Consequently the licence is widely deployed even though it is used by only about two percent of the open source-licensed projects on the software repository Sourceforge. It is an interesting licence to compare with the Berkeley Software Distribution (BSD) licence, which it resembles in some ways. This document attempts to draw together the main features of the Apache License into a friendly and comprehensible digest, and in addition to note some details about its history and usage.

History Of The Apache License

Beginning in 1995, the Apache Group (later the Apache Software Foundation) released successive versions of their well-known httpd server. Their initial licence was essentially the same as the old BSD licence, with only the names of the organisations changed. When Berkeley accepted the argument put to it by the Free Software Foundation and retired their *advertising clause* from the BSD licence, Apache did likewise and created the Apache License v1.1 - a slight variation on the *modified BSD licence*. In 2004 Apache decided to depart from the BSD model a little more radically, and produced the Apache License v2.

Main Features Of The Apache License V2

Like all licences the Apache License v2 grants certain rights under certain conditions. In brief a licensee of Apache Licensed V2 software can:

- copy, modify and distribute the covered software in source and/or binary forms
- exercise patent rights that would normally only extend to the licensor provided that:
- all copies, modified or unmodified, are accompanied by a copy of the licence
- all modifications are clearly marked as being the work of the modifier
- all notices of copyright, trademark and patent rights are reproduced accurately in distributed copies
- the licensee does not use any trademarks that belong to the licensor Furthermore, the grant of patent rights specifically is withdrawn if:
- the licensee starts legal action against the licensor(s) over patent infringements within the covered software This rewriting of the BSD licence tries to achieve a few things. Firstly it adds an explicit grant of patent rights where that is needed to operate, modify and distribute the software. Some argue that such a grant is implicit in other open source licences, but the Apache License v2 spells it out, as well as explicitly withdrawing that grant in the circumstance mentioned above. It also contains solid definitions of the concepts it uses, providing more certainty as to its intended meaning. Among these is a definition of *Contributor* that contains another interesting feature of the licence. A *Contributor*, as distinct from someone who just modifies the software, also grants a licence to their modification back to the original authors. This mechanism, if taken up, simplifies control of the code. Finally the v2 licence is usable by other projects without the need to replace wording in the licence document itself.

What is Linux explain?

Linux® is **an open source operating system (OS)**. An operating system is the software that directly manages a system's hardware and resources, like CPU, memory, and storage. The OS sits between applications and hardware and makes the connections between all of your software and the physical resources that do the work.

What is the advantage of Linux?

Linux provides many advantages over other operating systems: **Open-source software available for everyone to contribute, modify, and enhance the source code.** It is also available for users to download and use for free. Linux is less vulnerable and more secure than Windows operating systems.

What are features of Linux?

- Portable Environment. ...
- Free and Open-Source. ...
- Shell/ Command-line Interface. ...
- End-to-end encryption. ...
- Graphical User Interface (GUI) ...
- Configure Keyboards into Different Languages. ...
- Frequent New Updates. ...
- Lightweight Infrastructure.

What is Wikipedia

Wikipedia is a **free, open content online encyclopedia created through the collaborative effort of a community of users known as Wikipedians**. Anyone registered on the site can create an article for publication; registration is not required to edit articles.

JOOMLA

A Joomla component is **one of the main building blocks of your Joomla system**. When you just installed Joomla system on your website, it consists of the core, or, in other words, default components. If you wish to add some cool extra features to your default Joomla, you will have to download and install extra components.

Joomla Components

- FrontPage component - this is used to display the contents on the front page. ...
- WebLinks - this is a component where the user can configure different link categories and add items to each category.
- Banners - allows a Joomla website to create and display banners for advertising purposes.

What are advantages of components in Joomla?

Joomla's Benefits

- A flexible system, easy to extend and customise. ...
- Search engine optimised out of the box. ...
- Free forever Open Source software. ...
- Security conscious development with an excellent track record. ...
- Joomla! ...
- Grows organically as your needs evolve. ...
- Build powerful PHP applications.

Star Office

StarOffice is a **software product developed by Sun that includes a word processor, spreadsheet, presentation tool, drawing tool, and database**. StarDivision was founded in the 1980s and is the original developer of the StarOffice suite of software. This company was acquired by Sun Microsystems during the summer of 1999.

UNIT IV

Starting an open-source project

There are several ways in which work on an open-source project can start:

1. An individual who senses the need for a project announces the intent to develop a project in public.
2. A developer working on a limited but working codebase, releases it to the public as the first version of an open-source program.
3. The source code of a mature project is released to the public.
4. A well-established open-source project can be forked by an interested outside party.

Eric Raymond observed in his essay *The Cathedral and the Bazaar* that announcing the intent for a project is usually inferior to releasing a working project to the public.

It's a common mistake to start a project when contributing to an existing similar project would be more effective. To start a successful project it is very important to investigate what's already there. The process starts with a choice between the adopting of an existing project, or the starting of a new project. If a new project is started, the process goes to the Initiation phase. If an existing project is adopted, the process goes directly to the Execution phase.

Version Number

A version number is a unique number or set of numbers assigned to a specific release of a software program, file, firmware, device driver, or even hardware. Typically, as updates and entirely new editions of a program or driver are released, the version number will increase.

What is the meaning of the open source hardware?

"Open hardware," or "open source hardware," refers to the design specifications of a physical object which are licensed in such a way that said object can be studied, modified, created, and distributed by anyone. "Open hardware" is a set of design principles and legal practices, not a specific type of object.

What is open source designing?

Open-source design — is a design artifact (library, guide, font, code) that is published under an open source license, and is [usually] available on GitHub in the form of source files that can be modified freely. Examples: Google Fonts, few font foundries, Firefox Browser, Unsplash etc.

Licenses and Copyright for Free Hardware Designs

You make a hardware design free by releasing it under a free license. We recommend using the GNU General Public License, version 3 or later. We designed GPL version 3 with a view to such use.

Copyleft on circuits, and on nondecorative object shapes, doesn't go as far as one might suppose. The copyright on these designs only applies to the way the design is drawn or written. Copyleft is a way of using copyright law, so its effect carries only as far as copyright law carries.

For instance, a circuit, as a topology, cannot be copyrighted (and therefore cannot be copylefted). Definitions of circuits written in HDL can be copyrighted (and therefore copylefted), but the copyleft covers only the details of expression of the HDL code, not the circuit topology it generates. Likewise, a drawing or layout of a circuit can be copyrighted, so it can be copylefted, but this only covers the drawing or layout, not the circuit topology. Anyone can legally draw the same circuit topology in a different-looking way, or write a different HDL definition that produces the same circuit.

Copyright doesn't cover physical circuits, so when people build instances of the circuit, the design's license will have no legal effect on what they do with the devices they have built.

For drawings of objects, and 3D printer models, copyright doesn't cover making a different drawing of the same purely functional object shape. It also doesn't cover the functional physical objects made from the drawing. As far as copyright is concerned, everyone is free to make them and use them (and that's a freedom we need very much. All this may be true in your country as well, or it may not. Before producing objects commercially or in quantity, you should consult a local lawyer. Copyright is not the only issue you need to be concerned with. You might be attacked using patents, most likely held by entities that had nothing to do with making the design you're using, and there may be other legal issues as well.

What is open source in design?

Open Source Design is the development of technology and ideas without retention of intellectual property. The goal of the movement is to allow for the continued development and full customization of products. Typically this is done for software; however, it is increasingly being done for hardware.

Open Design principles.

Open Design embraces X centred design. Where X = the most critical factor of the project. This could be 'Human-centred design', 'Environment centred design' or 'Democracy centred design'

Open Design amplifies attitudes, methods, tools that are geared towards collaboration instead of competition, as well as a learning-based, mentoring approach.

Open Design encourages inclusion, low participation thresholds, and peer governance (free and communal validation of quality) thereby avoiding contributor burnout and design by committee.

Open Design is a multidisciplinary, intersectional and inclusive space.

Open Design advocates for transparency, accountability, and flexibility.

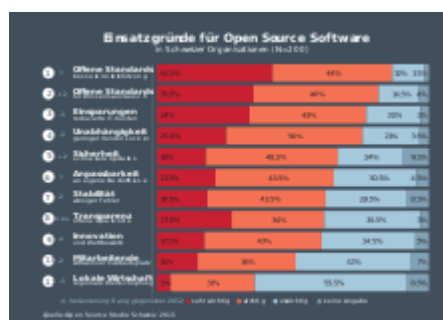
What is open source teaching?

Open-Source Learning **embraces the idea that everyone learns different things in different ways, and it values diverse approaches to reaching those goals.** Open-Source Learning empowers teachers and students to transform their ideas and interests into powerful learning experiences that are shared with everyone.

What is learning Object repository?

A learning object repository (LOR) is a **general term for an online collection of learning objects.** A learning object is an object or collection of objects that assists in the learning process (e.g., a topic, a module, a course, etc).

What are open source principles?



The open-source model is a decentralized software development model that encourages open collaboration. A main principle of open-source software development is **peer production**, with products such as source code, blueprints, and documentation freely available to the public.

Rules for Open Source Media

1. **Don't violate open source licenses.** If anyone tells you that open source licenses are never enforced, or violating them never causes problems, they are wrong. Violating open source licenses can cause legal liability and embarrassment, and slow down or compromise investments or acquisitions for your company. Even worse, it could make potential customers refuse to buy your products. Complying with software licenses is simply the right thing to do. Lots of developers have gone to great effort to make the software available to you; it's only fair to follow their licenses.
2. **Keep track of what you are using.** From day one, keep track of the open source software you are using. Someday you will have to provide a list, and having one already in hand will save you a lot of time and effort. Most open source software downloads have an included file called "license.txt". Keep a copy of the license file,

and note what software it covers, so you know what license terms you need to follow. Most startup companies do this with a simple spreadsheet.

3. **Understand Copyleft and Permissive licenses.** There are two kinds of open source licenses: copyleft and permissive. Copyleft licenses include GPL, LGPL, Eclipse Public License, Mozilla Public License, and the Common Development and Distribution License (CDDL). Permissive licenses include BSD, MIT, and Apache. There are many other licenses, but almost all open source software is under Copyleft or Permissive. Most companies — and their customers — have no legal concerns over using software under permissive licenses. Complying with Copyleft licenses takes more care, however, so if you want to use software under those licenses, be prepared to do what is necessary to comply.
4. **Comply with notice requirements.** All open source licenses have notice requirements. A typical notice requirement states that when you distribute open source software, you need to include a copy of the license. This usually means the entire license — not a link, not a short form. Notices can be challenging to maintain, particularly for embedded software. It's important to work out a license notice delivery strategy that complies with most open source licenses, but doesn't confuse or alienate your customers.
5. **Finger Pointing Doesn't Work.** If you violate an open source license, it doesn't matter if your supplier or contract developer was at fault. It's still your problem. Be sure all your suppliers and contractors follow the rules. For example, don't buy chips or computers with Linux if the supplier can't also provide you the source code.

UNIT 5

1. Open Source versus Closed Source

S.No.	OPEN SOURCE SOFTWARE	CLOSED SOURCE SOFTWARE
01.	Open source software refers to the computer software which source is open means the general public can access and use.	Closed source software refers to the computer software which source code is closed means public is not given access to the source code.
02.	Open Source Software in short also referred as OSS.	Closed Source Software in short also referred as CSS.
03.	The source code of open source software is public.	In closed source software the source code is protected.
04.	This code can be modified by other users and organizations means that the source code is available for anyone to look at.	The only individual or organization who has created the software can only modify the code.
05.	The price of open source software is very less.	The price of closed source software is high.
06.	There is no so much restrictions on users based on usability and modification of software.	There is so much restrictions on users based on usability and modification of software.
07.	Programmers compete with each other for recognition.	Programmers do not compete with each other for recognition.
08.	Programmers freely provide improvement for recognition if their improvement is accepted.	Programmers are hired by the software firm/organization to improve the software.
09.	If the program is popular then very large number of programmers may work on the project.	There is a limitation on the number of programmers/team who will work on the project.
10.	It is purchased with its source code.	

11.	Open software can be installed into any computer.	Closed software needs have a valid license before installation into any computer.
12.	Open source software fails fast and fix faster.	Closed source software has no room for failure.
13.	In closed source software no one is responsible for the software.	In closed source software the vendor is responsible if anything happened to software.
14.	Examples are Firefox, OpenOffice, Gimp, Alfresco, Android, Zimbra, Thunderbird, MySQL, Mailman, Moodle, TeX, Samba, Perl, PHP, KDE etc.	Examples are Skype, Google earth, Java, Adobe Flash, Virtual Box, Adobe Reader, Microsoft office, Microsoft Windows, WinRAR, mac OS, Adobe Flash Player etc

Open Source

Richard Stallman's organization is called the free software foundation, not the open source software foundation. There's a reference to "FSF's mission to preserve protect and promote the freedom to use, study, copy, modify and redistribute computer software".

Open Source Government

Primarily refers to use of open source software technologies in traditional government organization and government operations such as voting.

Open politics

It is also known as Open source Politics. It's a term used to describe a political process that uses Internet technologies such as blogs, email and polling to provide for a rapid feedback mechanism between political organization of the open source politics which relates to the development of public under a set of rules and processes similar to the open source software movement.

The ethics of Open Source

Open source ethics is described from the three perspectives of three key roles:

- Commercial Consumer & Redistributors
- Open Source Provider
- Open Source Contributor

Shared software:

Shared software is a different term used to describe free software open source software and possibly also software that is not formally covered by the definition of either, but that is in some other way shared rather than owned.

Shared Source

Shared Source is Microsoft framework for sharing computer program source code with individuals and organizations. Microsoft's shared source initiative includes a spectrum of technologies and licenses. Most of its offerings are available for download by anyone.

commercialization of open source

There are two common ways OSS is commercialized. The easiest (and most obvious) way to monetize is **consulting or support**. Since you created the tech, customers will assume you know it the best. The ideal way to monetize open source is by building a commercial software extension to your free offering.

What is open source government in software?

Open source means **the source code is available for access and customization**. Government agencies can easily tailor open source applications to meet their specific needs and requirements and enjoy complete ownership of their application. A common misconception among people is that open source implies free of cost.

What is ethics of open source?

A method and philosophy for software licensing and distribution designed to encourage use and improvement of software written by volunteers by ensuring that anyone can copy the source code and modify it freely.

Open source

Software that is distributed with its source code so that end user organizations and vendors can modify it for their own purposes. Most open source licenses allow the software to be redistributed without restriction under the same terms of the license.

There are thousands of open source titles, and although used on most platforms, they are particularly common in the Linux/Unix world. Major examples are the Linux operating system, Apache Web server and JBoss application server. See **Linux**, **Apache** and **JBoss**.

Free and Paid

A great amount of open source software is available at no charge, and many open source proj

ects are developed by a community of volunteers. However, there are commercial vendors that enhance open source software, provide technical support and charge a fee, the most notable example being the Linux operating system.

Fewer Bugs vs. Too Many Cooks

Open source proponents claim that the code in open source programs is continuously reviewed by new programmers, and bugs are weeded out more quickly. In addition, the wide variety of contributors leads to enhancements that might not otherwise be included. However, vendors of proprietary software counter by saying that having total control over the software is far superior; that the old saying "too many cooks spoil the broth" is especially true in software development.

Advantages of Open Source

The single major advantage is no licensing fees. A second advantage is flexibility. Users of open source software can modify the programs to fit their needs, and many companies use open source for this reason alone.

Another advantage is the ability to fast track a project. Since there is no contract to sign, IT departments can use the software without waiting for contractual negotiations to be completed.

Lastly, as long as there is one remaining, devoted contributor, the software will continue to be enhanced. In the world of proprietary, commercial software, a useful program that users may truly love is often abandoned when it no longer generates enough profit.

The Free and Open Source Communities

In the late 1990s, open source software was derived from "free software," meaning free of restrictions and why the phrase "free and open source software" is often used. Whereas the "free software" movement promotes the user's freedom as an ethical issue, the philosophy of open source focuses on the practical benefits when users cooperate with each other.